

Decent Coaching Classes
VIII SB
Academic Conquest - IV
Subject: Geography — ANSWER KEY

Q.1. Choose the correct alternative:

1. (b) mantle
2. (b) 7 to 10
3. (b) Moho
4. (b) absolute humidity
5. (c) high
6. (b) continental shelf
7. (a) Mariana Trench
8. (b) thermohaline

Q.2. Fill in the blanks:

1. 2900
2. Secondary
3. condensation
4. Cumulonimbus
5. 3700
6. thermohaline

Q.3. True or False (corrected statements):

1. False — The outer core of the earth's interior is liquid/semi-liquid in nature (only the inner core is solid).
2. True.
3. False — The continental crust is made up of silica and aluminium (the oceanic crust is made up of silica and magnesium).
4. False — The continental shelf is the shallowest part of the ocean floor.
5. False — Warm ocean currents make the climate of the nearby coast warmer.
6. True.

Q.4. Geographical reasons:

1. As we move from the earth's surface to its centre, major changes occur in temperature and density. The crust, mantle and core differ in composition, density and physical state (solid/liquid), which creates variety in the earth's interior.
2. The internal energy released due to endogenetic movements occurring in the mantle (especially the asthenosphere, where epicentres of deep-seated earthquakes are found and magma chambers exist) is responsible for mountain-building, rifts, volcanic eruptions and earthquakes.
3. Clouds are formed of condensed water or snow particles which are very fine and almost weightless. Because of this and the vertical flow of wind, clouds float in the atmosphere instead of falling.
4. The continental shelf is shallow, so sunlight reaches its bed, allowing algae and plankton to grow, which is food for fish. It also yields natural gas and mineral oil, but its main importance for fishing is the abundant food supply it provides for marine life.
5. Marine islands are the summits of submerged hills and mountains on the ocean floor that rise above sea level. Since the rest of the mountain lies underwater, only its peak is visible as an island (e.g., Iceland, Andaman and Nicobar Islands).
6. The eastern coast of Canada does not receive the warming influence of a warm ocean current (unlike Western Europe, which is warmed by the Gulf Stream). In the absence of a warm current, cold conditions dominate and the ports freeze in winter.

Q.5. Distinguish between:**1. Continental crust and Oceanic crust:**

Continental crust is made up of silica and aluminium (earlier called Sial), has a density of 2.65 to 2.90 gm/cm³, is about 30 km thick, and mainly contains granite rocks. Oceanic crust is made up of silica and magnesium (earlier called Sima), has a density of 2.9 to 3.3 gm/cm³, is about 7 to 10 km thick, and mainly contains basalt and gabbro rocks.

2. Relative humidity and Absolute humidity:

Absolute humidity is the actual amount of water vapour present in 1 cu.m. of air, expressed in grams.

Relative humidity is the amount of water vapour present in the air expressed as a percentage of the amount needed for saturation at the same temperature and volume.

3. Continental shelf and Continental slope:

The continental shelf is the shallow, gently-sloping part of the ocean floor near the coast, extending up to a depth of 200 m; it is rich in fishing grounds, oil and gas. The continental slope lies beyond the shelf, has a much steeper gradient, extends from 200 m to about 3600 m, and its lower boundary marks the boundary of the continent.

Q.6. Answer in detail:

1. Diagram: The interior of the earth is shown as concentric layers — Crust (outermost, further divided into continental and oceanic crust), Mantle (divided into upper mantle/asthenosphere and lower mantle), and Core (divided into outer core and inner core), with the inner core at the earth's centre. Description: The Crust is the thinnest, solid outer layer (continental crust: silica+aluminium, ~30 km thick; oceanic crust: silica+magnesium, ~7-10 km thick). The Mantle lies below the crust, extends to about 2900 km depth, and its upper part (asthenosphere) is semi-liquid with magma chambers, while the lower mantle is denser. The Core extends from about 2900 km to the earth's centre (6371 km); the Outer Core (2900-5100 km) is liquid, generating the earth's magnetic field, while the Inner Core (5150-6371 km) is solid due to extreme pressure, made mainly of iron and nickel.

2. Clouds are classified into three main types on the basis of their altitude: High Clouds form at 7000-14000 m and are made of fine snow crystals; they include cirrus (wispy strands), cirrocumulus (small wave-like groups) and cirrostratus (sheet-like, with a halo). Medium Clouds form at 2000-7000 m and include alto-cumulus (layered, wavy, white with grey shade) and alto-stratus (thin, sun visible through them like a milky glass). Low Clouds form below 2000 m and include strato-cumulus, stratus (ash-coloured, layered), nimbostratus (thick, grey, causes continuous rainfall), cumulus (dome-shaped, indicate pleasant weather) and cumulonimbus (huge, dark, mountain-like, bring thunderstorms, lightning and hail).